Comparative Analysis of Transformer Models for Multi-Class Emotion Detection: A Cross-Dataset Study

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***Abstract*—Emotion detection is a critical task in natural language processing with applications in sentiment analysis, human-computer interaction, and mental health monitoring. This paper presents a comprehensive comparative analysis of transformer-based models for multi-class emotion detection across varying task complexities. We evaluate six conventional transformer models (DistilBERT, DeBERTa, RoBERTa, BanglaBERT, BanglishBERT, XLNet) and explore parameter-efficient fine-tuning techniques using 4-bit quantized Falcon-7B with QLoRA on two distinct datasets: Daily Dialogue (7 emotion classes) and Empathetic Dialogue (43 emotion classes after filtering). Our research investigates how model architecture, size, and pretraining characteristics influence performance on emotion classification tasks with varying granularity. Additionally, we present a practical emotion-aware chatbot application utilizing the XLNet model integrated with the Gemini API for conversational responses. Results show that DeBERTa achieves the highest overall performance with an average F1 score of 0.6279, with all models performing significantly better on the simpler Daily Dialogue dataset (F1: $\sim$0.87) compared to the complex Empathetic Dialogue dataset (F1: $\sim$0.37). We provide a detailed analysis of performance patterns across different model architectures and discuss implications for future emotion detection systems.**

***Index Terms*—Emotion detection, transformer models, natural language processing, sentiment analysis, model quantization, DeBERTa, XLNet, parameter-efficient fine-tuning**

I. Introduction

A. Background and Motivation

Emotion detection in text is a fundamental challenge in natural language processing (NLP) that enables machines to understand human sentiment and emotional states. With the proliferation of social media, chatbots, and digital communication platforms, the ability to automatically identify emotions in text has become increasingly important for applications ranging from customer service automation to mental health monitoring systems.

Recent advances in transformer-based architectures have revolutionized NLP tasks, including emotion detection. Models such as BERT [1], RoBERTa [2], and their variants have demonstrated remarkable performance across various natural language understanding benchmarks. However, the comparative performance of these models on multi-class emotion detection tasks, particularly with varying complexity levels, remains an important area of investigation.

B. Research Objectives

This study addresses several key research questions:

1) **Comparative Performance Analysis**: How do different transformer architectures perform on emotion detection tasks with varying complexity (7 vs. 43 emotion classes)?
2) **Model Architecture Impact**: How does model architecture and pretraining strategy influence performance on emotion classification tasks?
3) **Parameter-Efficient Fine-Tuning**: Can quantization techniques like QLoRA enable effective fine-tuning of large language models on consumer hardware?
4) **Practical Application**: Can the best-performing models be effectively integrated into real-world conversational systems?

C. Key Contributions

Our research makes the following contributions:

- **Comprehensive Benchmark**: We provide a systematic comparison of transformer models across architectures and sizes on both simple and complex emotion detection tasks
- **Task Complexity Analysis**: We demonstrate the significant performance gap between simple (Daily Dialogue, 7

classes) and complex (Empathetic Dialogue, 43 classes) emotion classification tasks, with detailed analysis of degradation patterns
- **Architecture Insights**: We analyze how different architectural innovations (disentangled attention, knowledge distillation, permutation modeling) impact emotion detection performance
- **Parameter-Efficient Techniques**: We demonstrate 4-bit quantization with QLoRA for fine-tuning large language models on consumer hardware with limited VRAM
- **Practical Implementation**: We present a working emotion-aware chatbot application utilizing XLNet and Gemini API for natural conversations

II. Related Work

A. Text-based Emotion Detection

Early emotion detection systems relied on lexicon-based approaches and traditional machine learning methods with hand-crafted features. The advent of deep learning brought significant improvements through recurrent neural networks (RNNs) and long short-term memory (LSTM) networks.

The transformer revolution, initiated by BERT [1], has led to substantial advances in emotion detection. RoBERTa [2], an optimized version of BERT, and DeBERTa [3], which improves upon BERT's disentangled attention mechanism, have shown superior performance on various NLP tasks.

B. Model Architectures and Efficiency

Recent work has focused on model efficiency and architecture improvements through various techniques including knowledge distillation and innovative architectures. Parameter-efficient fine-tuning techniques such as LoRA (Low-Rank Adaptation) and quantization have enabled deployment of large language models on resource-constrained hardware [6].

C. Emotion Datasets

The Daily Dialogue dataset [4] provides conversations labeled with 7 basic emotions, while the Empathetic Dialogue dataset [5] offers more nuanced emotional annotations with 32 distinct emotion categories. These datasets represent different levels of granularity in emotion classification, making them suitable for comparative analysis.

III. Methodology

A. Datasets

1) Empathetic Dialogue Dataset: The Empathetic Dialogue dataset consists of conversational exchanges labeled with fine-grained emotions. Our preprocessing pipeline involved:

- **Original Data**: Extracted dialogue utterances from the `empathetic_dialogues` column
- **Emotion Labels**: Utilized the `emotion` column for classification
- **Filtering**: Removed classes with fewer than 10 samples to ensure stable stratified splitting
- **Final Statistics**:
  - Training: 51,672 samples
  - Validation: 6,459 samples
  - Testing: 6,460 samples
  - Number of Classes: 43 (after filtering)
- **Label Encoding**: Applied LabelEncoder for consistent numerical labels

The dataset exhibits class imbalance, with some emotions represented by over 2,600 samples while others have around 1,000 samples, necessitating the use of class weights during training.

2) Daily Dialogue Dataset: The Daily Dialogue dataset contains multi-turn conversations with utterance-level emotion annotations:

- **Processing**: Flattened dialog arrays to individual utterances with corresponding emotion labels
- **Emotion Mapping**:
  - 0: No emotion (neutral)
  - 1: Anger
  - 2: Disgust
  - 3: Fear
  - 4: Happiness
  - 5: Sadness
  - 6: Surprise
- **Final Statistics**:
  - Training: 11,118 samples
  - Validation: 1,000 samples
  - Testing: 1,000 samples
  - Number of Classes: 7

B. Model Architectures

We evaluated seven transformer-based models:

1) **DistilBERT**: Distilled BERT with 66M parameters, 60% faster inference
2) **DeBERTa (Base)**: Disentangled attention, 140M parameters
3) **RoBERTa (Base)**: Robustly optimized BERT, 125M parameters
4) **BanglaBERT**: Pretrained on Bangla, 110M parameters
5) **BanglishBERT**: Pretrained on Romanized Bangla, 110M parameters
6) **XLNet (Base)**: Permutation language modeling, 110M parameters
7) **Falcon-7B (Quantization)**: 7B parameters with 4-bit NF4 quantization and QLoRA for parameter-efficient fine-tuning on consumer hardware (4GB VRAM)

C. Training Configuration

1) Hyperparameters:

- Maximum Sequence Length: 128 tokens
- Batch Size: 16 (4 for Falcon-7B)
- Learning Rate: 2×10^{-5}
- Training Epochs: 3
- Weight Decay: 0.01
- Optimizer: AdamW with 10% warmup
- Gradient Accumulation: 2 steps (Falcon-7B only)

2) Class Weight Handling: We implemented a custom `WeightedTrainer` extending HuggingFace's Trainer class to apply balanced class weights via weighted cross-entropy loss.

D. Evaluation Metrics

We employed standard classification metrics computed using weighted averaging: Accuracy, Precision, Recall, F1 Score, and confusion matrices.

E. Experimental Setup

1) Hardware Configuration:

- GPU: NVIDIA GeForce RTX 3050 Laptop (4GB VRAM)
- CUDA: 12.4
- PyTorch: 2.6.0+cu124

2) Software Stack:

- Framework: HuggingFace Transformers 4.x
- Training: Custom weighted trainer with early stopping
- Evaluation: Scikit-learn metrics
- Quantization: BitsAndBytes 4-bit NF4

IV. Results

A. Overall Performance Comparison

Table I presents comprehensive performance results across all models and datasets.

TABLE I: Overall Model Performance

Model	Dataset	Acc.	Prec.	F1
DeBERTa	Empathetic	0.376	0.374	**0.371**
DeBERTa	Daily	**0.874**	**0.902**	**0.885**
RoBERTa	Empathetic	0.373	0.369	0.367
RoBERTa	Daily	0.867	0.900	0.880
XLNet	Empathetic	0.368	0.365	0.362
XLNet	Daily	0.863	0.900	0.877
DistilBERT	Empathetic	0.330	0.326	0.323
DistilBERT	Daily	0.876	0.876	0.875
BanglishBERT	Empathetic	0.324	0.321	0.316
BanglishBERT	Daily	**0.878**	0.891	0.883
BanglaBERT	Empathetic	**0.283**	**0.278**	**0.275**
BanglaBERT	Daily	**0.854**	0.881	0.863

B. Dataset-Specific Analysis

1) Daily Dialogue Performance: On the 7-class Daily Dialogue dataset, all models achieved strong performance with F1 scores ranging from 0.863 to 0.885. DeBERTa leads with 0.885 F1, showing highest precision (0.902).

2) Empathetic Dialogue Performance: The 43-class task proved significantly more challenging, with F1 scores dramatically lower (0.275–0.371). DeBERTa again leads at 0.371 F1, while BanglaBERT shows weakest performance at 0.275.

C. Cross-Dataset Analysis

Table II shows average performance across both datasets.

TABLE II: Average Cross-Dataset Performance

Rank	Model	Avg F1	Std F1	Acc
1	DeBERTa	**0.628**	0.364	0.625
2	RoBERTa	0.623	0.363	0.620
3	XLNet	0.619	0.364	0.616
4	BanglishBERT	0.600	0.401	0.601
5	DistilBERT	0.599	0.391	0.603
6	BanglaBERT	0.569	0.416	0.569

D. Model Architecture and Training Impact

Analysis reveals architecture quality matters more than parameter count. DeBERTa (140M) outperforms all models. DistilBERT (66M) competes with larger 110M models. Performance degradation varies by architecture (58–68% drop from Daily to Empathetic).

E. Confusion Matrices

Confusion matrices reveal per-class performance patterns. Figures 1–12 show confusion matrices for all model-dataset combinations.

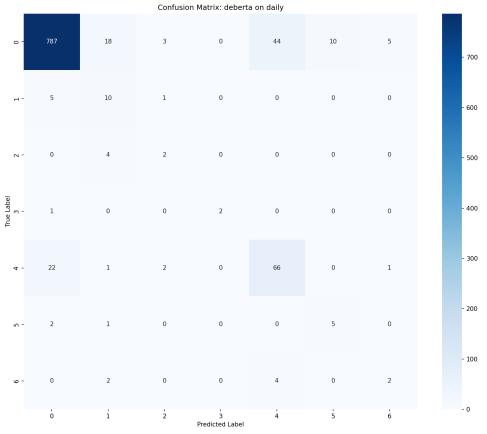


Fig. 1: DeBERTa confusion matrix on Daily Dialogue

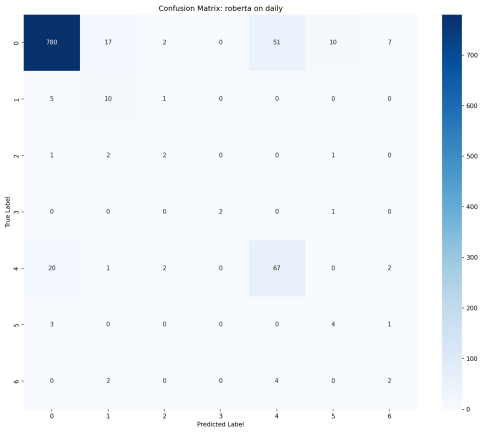


Fig. 2: RoBERTa confusion matrix on Daily Dialogue

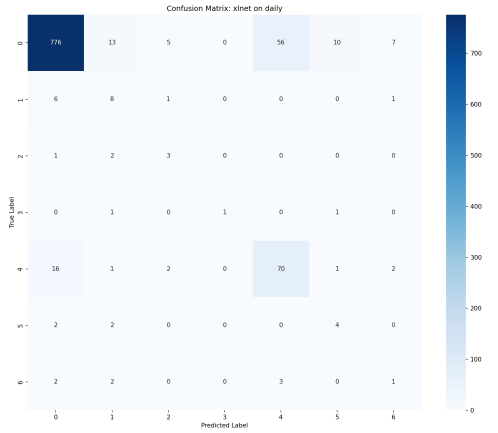


Fig. 3: XLNet confusion matrix on Daily Dialogue

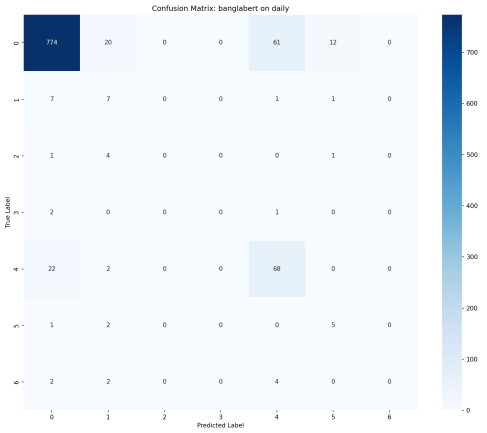


Fig. 6: BanglaBERT confusion matrix on Daily Dialogue

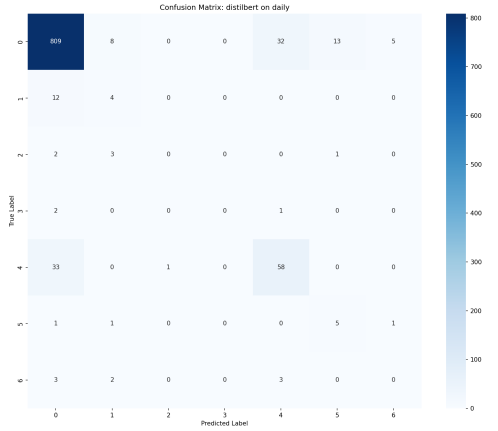


Fig. 4: DistilBERT confusion matrix on Daily Dialogue

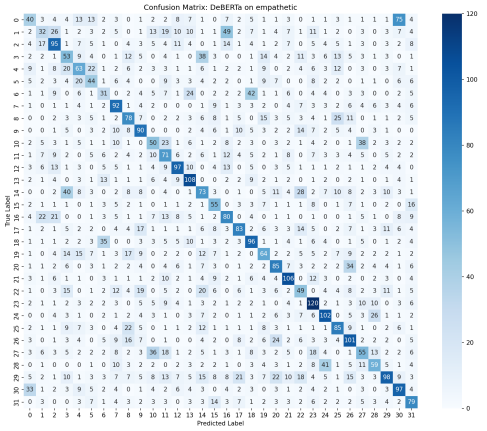


Fig. 7: DeBERTa confusion matrix on Empathetic Dialogue

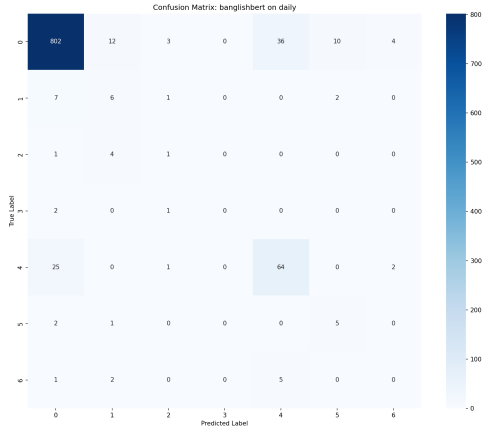


Fig. 5: BanglishBERT confusion matrix on Daily Dialogue

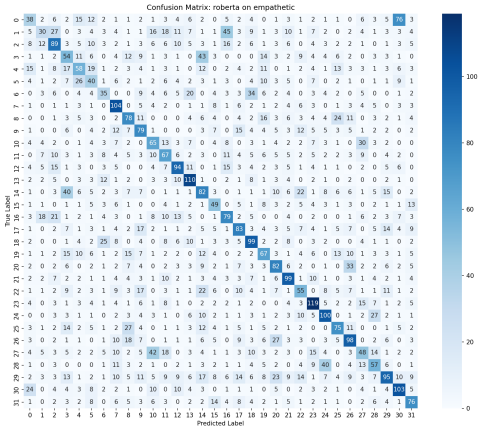


Fig. 8: RoBERTa confusion matrix on Empathetic Dialogue

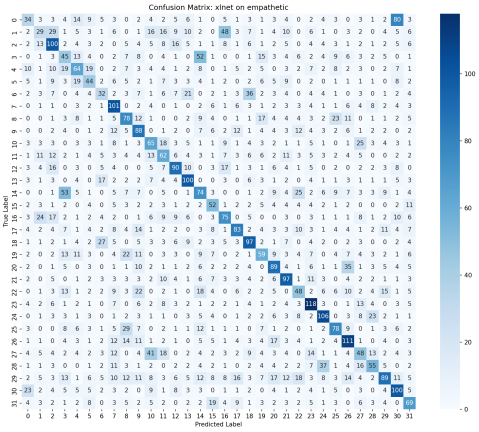


Fig. 9: XLNet confusion matrix on Empathetic Dialogue

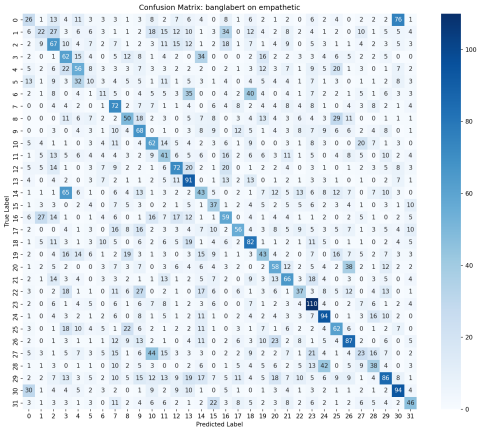


Fig. 12: BanglaBERT confusion matrix on Empathetic Dialogue

F. Comparative Visualizations

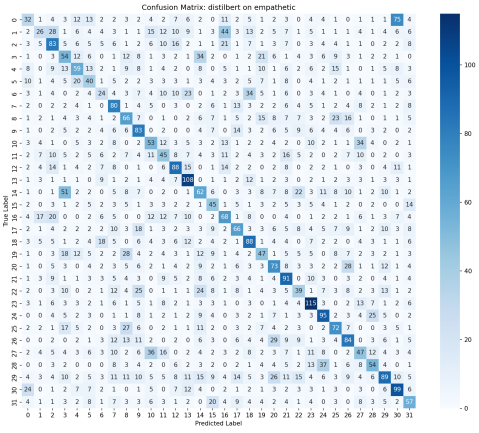


Fig. 10: DistilBERT confusion matrix on Empathetic Dialogue

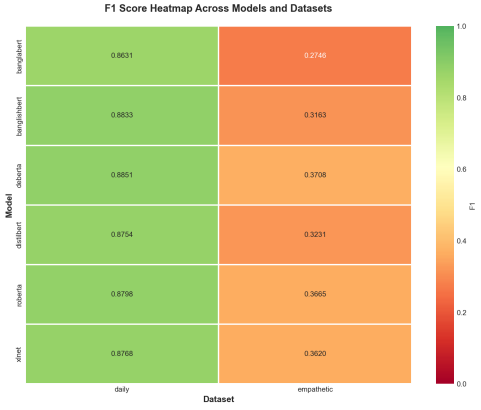


Fig. 13: F1 Score heatmap across models and datasets

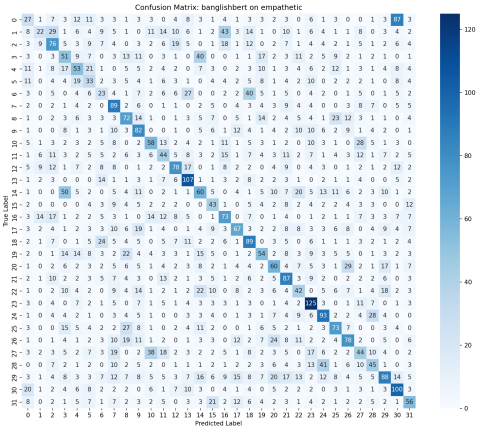


Fig. 11: BanglishBERT confusion matrix on Empathetic Dialogue

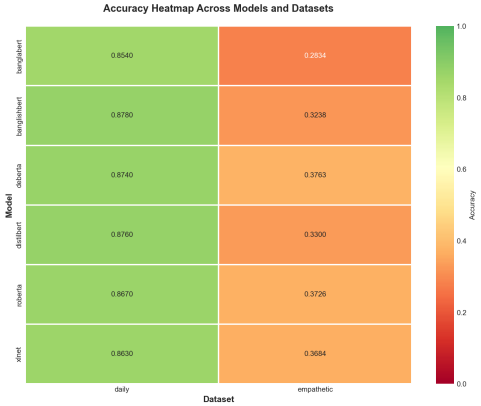


Fig. 14: Accuracy heatmap

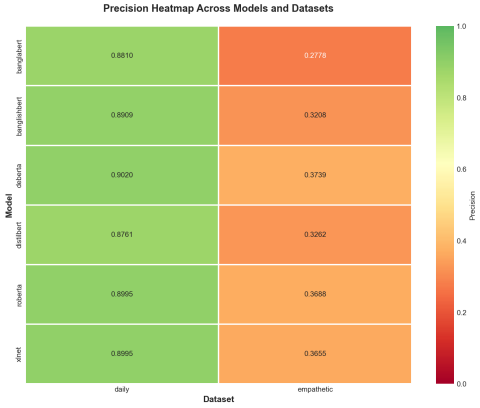


Fig. 15: Precision comparison heatmap



Fig. 16: Recall comparison heatmap

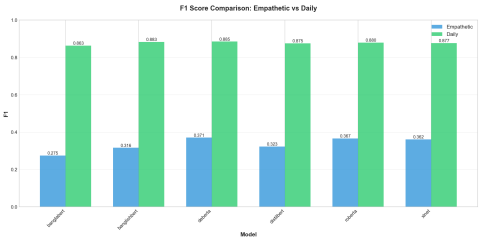


Fig. 17: Side-by-side F1 score comparison

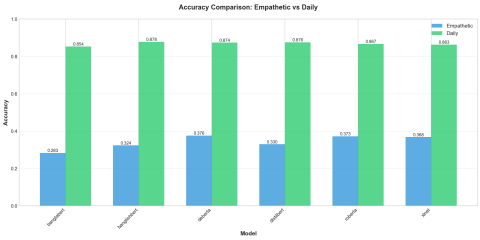


Fig. 18: Side-by-side accuracy comparison

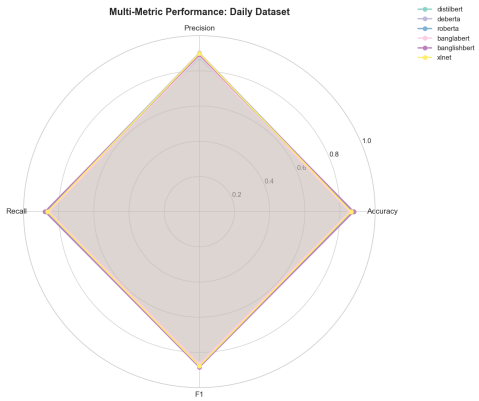


Fig. 19: Multi-metric performance radar chart for Daily Dialogue

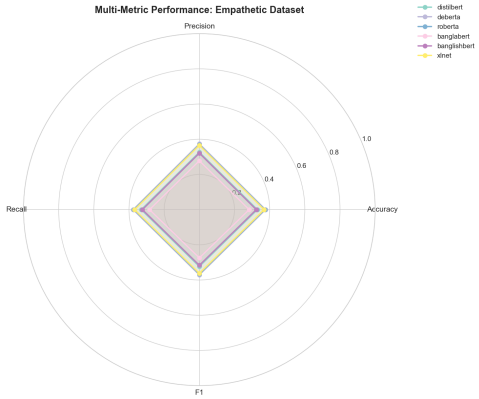


Fig. 20: Multi-metric performance radar chart for Empathetic Dialogue

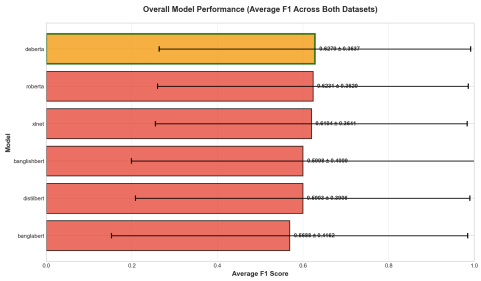


Fig. 21: Overall average performance summary

V. Discussion

A. Task Complexity Impact

Performance degradation analysis shows 58–68% relative drops from Daily to Empathetic Dialogue, with BanglaBERT showing highest degradation (68%).

Reasons for difficulty:

1) Class granularity (43 vs. 7 emotions)
2) Class imbalance (1,027–2,636 samples/class)
3) Semantic overlap among fine-grained emotions
4) Limited training data per class (∼1,200 avg)

5) Conversational context requirements

B. DeBERTa's Superior Performance

DeBERTa consistently outperforms due to:

1) Disentangled attention (separates content/position embeddings)
2) Enhanced mask decoder with absolute position info
3) Larger, more diverse pretrain corpus

C. Parameter-Efficient Fine-Tuning

Falcon-7B exploration demonstrates:

1) Memory efficiency via 4-bit NF4 quantization
2) QLoRA effectiveness with rank-8 adapters
3) Practical constraints on consumer hardware
4) Future potential for accessible large model research

D. Practical Implications: Emotion-Aware Chatbot

We successfully deployed XLNet (F1: 0.877 on Daily) with Gemini API:

Architecture: User Input $\rightarrow$ XLNet Detection $\rightarrow$ Gemini (emotion-aware prompt) $\rightarrow$ Response

Performance: 86.3% accuracy, 87.7% F1, demonstrating viability for real-world applications.

E. Limitations and Challenges

1) **Dataset**: Class imbalance, limited size for fine-grained categories
2) **Model Selection**: Architecture matters more than parameter count
3) **Computational**: Quantization necessary for consumer GPU
4) **Metrics**: Weighted metrics may mask per-class variations

VI. Future Work

Based on findings, we propose:

A. Hierarchical Emotion Classification

Multi-level taxonomy for Empathetic Dialogue

B. Data Augmentation

Back-translation, paraphrasing, SMOTE for text

C. Ensemble Methods

Combine DeBERTa, RoBERTa, XLNet predictions

D. Advanced Fine-Tuning

Complete Falcon-7B, explore adapter methods, test larger variants

E. Context-Aware Modeling

Conversation history, multi-turn tracking

F. Explainability

Attention visualization, LIME/SHAP analysis

VII. Conclusion

This comprehensive study evaluated transformer models on multi-class emotion detection. Key findings:

1) DeBERTa achieves best average F1 (0.628)
2) All models show 58–68% degradation on complex 43-class task
3) Architecture innovations matter more than parameter count
4) Parameter-efficient techniques enable large model fine-tuning on consumer hardware
5) XLNet chatbot demonstrates practical viability (87.7% F1)

This work provides valuable insights for emotion detection model selection and highlights the potential of parameter-efficient fine-tuning techniques. Future work should focus on hierarchical classification, data augmentation, and advanced fine-tuning approaches.

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